SOFTWARE TESTING ASSIGNMENT - PART 5

WBS, Test Effort Estimation, and Test Point Analysis (TPA)

> **1. Pick any one feature from an app you use daily (for example: 'Search Restaurants' in Zomato or 'Add to Cart' in Flipkart) and break down the testing tasks for this feature using a Work Breakdown Structure (WBS) — list at least 6 distinct testing activities or sub-tasks.**

Feature: Add to Cart (Flipkart-like App)

WBS Detailed Activities

1. **Requirement Analysis:** Review requirements for the Add to Cart feature and identify business rules and expected behavior.
2. **Test Planning:** Define testing scope, approach, and prepare high-level test scenarios and detailed test cases.
3. **Functional Testing:** Verify single and multiple products can be added, and confirm the cart count updates correctly.
4. **Validation Testing:** Test handling of out-of-stock products, adding identical items, and restriction of quantity limits.
5. **UI Testing:** Verify Add to Cart button visibility, alignment, responsivity, and text/icon rendering in the cart layout.
6. **Integration Testing:** Verify cart data persistence in the database and multi-device account synchronization after login.
7. **Performance Testing:** Validate quick checkout responsiveness and cart loading benchmarks under a heavy simulated user load.
8. **Defect Reporting & Retesting:** Log discovered bugs, coordinate with developers, and retest bug fixes alongside regression checks.

WBS Visual Diagram Structure

```
Add to Cart Feature Testing
│
├── 1. Requirement Analysis
├── 2. Test Planning
├── 3. Functional Testing
├── 4. Validation Testing
├── 5. UI Testing
├── 6. Integration Testing
├── 7. Performance Testing
└── 8. Defect Reporting & Retesting
```

> **2. For the 'Send Message' feature in WhatsApp, estimate the total test effort required by listing all the main test activities, assigning estimated hours to each, and calculating the overall effort needed.**

WhatsApp 'Send Message' Effort Estimation

Activity	Description	Estimated Hours
1. Requirement Analysis	Review feature requirements and understand message status delivery flows.	2 Hours
2. Test Case Design	Create specific test scenarios and detailed logical test cases.	4 Hours
3. Test Environment Setup	Prepare test devices, target accounts, and varying network environments.	2 Hours
4. Test Execution	Execute all functional, multi-media, and negative input test cases.	6 Hours
5. Defect Reporting	Log and document code defects and bugs found during execution steps.	2 Hours
6. Defect Retesting	Verify that reported bugs are successfully fixed by the development team.	2 Hours
7. Regression Testing	Ensure hotfixes and patches have not broken existing core module functions.	3 Hours
8. Test Closure	Prepare final test summary closure documentation and update dashboard statuses.	1 Hour
Total Estimated Effort Needed		**22 Hours**

> **3. Given the following scenario: Flipkart's 'Apply Coupon' feature is being released as a small module. Using the steps of test effort estimation discussed in class, create a simple estimation sheet (table) showing each test activity, estimated hours, and any assumptions you make.**

Test Effort Estimation Sheet – Flipkart "Apply Coupon" Feature

Core Structural Assumptions:

- The target feature module scope is structurally small.
- Coupon code validation core engines are already built and active.
- A dedicated staging test environment is stable and ready.
- A set of around 10–15 unique target test cases needs execution cycles.
- A single qualified QA tester is allocated full-time.

Estimation Breakdown

Test Activity	Description	Estimated Hours
Requirement Analysis	Review coupon system business rules and logic boundaries.	1 Hour
Test Planning	Identify product testing scope, dependency risks, and approaches.	1 Hour
Test Case Design	Build test cases for valid, invalid, expired, and multi-tier coupon types.	3 Hours
Test Data Prep	Generate valid/invalid coupon promotional codes and customer accounts.	1 Hour
Test Execution	Run all planned test scripts against checkout interfaces.	4 Hours
Defect Reporting	Log discrepancies, boundary edge errors, and interface issues.	1 Hour
Defect Retesting	Verify code resolutions for rejected coupon logs.	2 Hours
Regression Testing	Validate that coupon system modifications do not interrupt payment transactions.	2 Hours
Test Summary Report	Draft product sign-off and metrics summaries.	1 Hour
Total Estimated Effort		**16 Hours**

4. Read about Test Point Analysis (TPA) as introduced in class. In your own words, explain how TPA could help estimate testing effort for a new 'Movie Booking' feature in BookMyShow. List two advantages and one limitation of using TPA for this purpose.

Test Point Analysis (TPA) Evaluation (BookMyShow Movie Booking)

Concept Application: Test Point Analysis (TPA) is a structured approach used to evaluate the size and functional footprint of a software application feature to establish precise effort metrics rather than relying on ad-hoc estimation. For a new *Movie Booking* feature on BookMyShow, TPA systematically weights sub-components such as movie searching, theater/showtime selection, graphic interactive seat layouts, transaction processors, and ticket delivery modules. Elements with complex business rules or intensive backend interactions are assigned higher Test Points, mapping directly to time, cost, and manpower requirements.

Advantages of TPA

- **Highly Objective and Accurate:** Eliminates guesswork by mapping estimations directly to tangible component parameters, data elements, and complexity metrics.
- **Optimized Resource Scheduling:** Allows QA leads to mathematically predict team allocations and pinpoint precise delivery dates based on point metrics.

Limitation of TPA

- **Dependence on Immutable Requirements:** TPA is highly dependent on mature, complete, and frozen product specifications. If the movie booking features face shifting requirements or fluid logic halfway through development, initial point structures crumble, causing inaccurate estimates.